STAT 153 & 248 - Time Series
Lecture Eighteen

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1 Predictions from $AR(p)$ models

If the parameters $\phi_0, \ldots, \phi_p$ and σ (collectively denoted by θ) of the AR(p) model are fixed, then the prediction for a future value y_{n+i} is given by

$$\hat{y}_{n+i}(\theta) := \mathbb{E}\left(y_{n+i} \mid y_1, \ldots, y_n, \theta\right)$$

These values are calculated recursively for $i = 1, 2, \ldots$ using the following recursion:

$$\hat{y}_{n+i}(\theta) = \phi_0 + \phi_1 \hat{y}_{n+i-1}(\theta) + \phi_2 \hat{y}_{n+i-2}(\theta) + \cdots + \phi_p \hat{y}_{n+i-p}(\theta) \qquad \text{for } i = 1, 2, \ldots. \tag{1}$$

which is initialized by

$$\hat{y}_j(\theta) = y_j \qquad \text{for } j = n, n-1, \ldots, n+1-p, \tag{2}$$

Since θ is unknown, we can replace it by the conditional MLE $\hat{\theta}$.

2 Prediction Standard Errors

To compute prediction standard errors, we can again fix the parameter values θ, and then attempt to calculate:

$$V_i(\theta) := \operatorname{var}\left(y_{n+i} \mid y_1, \ldots, y_n, \theta\right) \qquad \text{for } i = 1, 2, \ldots.$$

The prediction standard error corresponding to the predicted value for y_{n+i} can then be taken to be $\sqrt{V_i(\hat{\theta})}$ (note that θ is replaced by the conditional MLE $\hat{\theta}$).

It turns out that it is not really possible to setup a recursion for $V_i(\theta)$. To see this, note that:

$$\begin{aligned} V_1(\theta) &= \operatorname{var}(y_{n+1} \mid y_1, \ldots, y_n, \theta) \\ &= \operatorname{var}(\phi_0 + \phi_1 y_n + \phi_2 y_{n-1} + \cdots + \phi_p y_{n+1-p} + \epsilon_{n+1} \mid y_1, \ldots, y_n, \theta) \\ &= \operatorname{var}(\epsilon_{n+1} \mid y_1, \ldots, y_n, \theta) = \sigma^2. \end{aligned}$$

Next

$$\begin{aligned}
V_2(\theta) &= \text{var}(y_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \text{var}(\phi_0 + \phi_1 y_{n+1} + \phi_2 y_n + \cdots + \phi_p y_{n+2-p} + \epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \text{var}(\phi_1 y_{n+1} + \epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \phi_1^2 \text{var}(y_{n+1} \mid y_1, \ldots, y_n, \theta) + \text{var}(\epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \phi_1^2 V_1(\theta) + \sigma^2 = \sigma^2(1 + \phi_1^2).
\end{aligned}$$

The next term is

$$\begin{aligned}
V_3(\theta) &= \text{var}(y_{n+3} \mid y_1, \ldots, y_n, \theta) \\
&= \text{var}(\phi_0 + \phi_1 y_{n+2} + \phi_2 y_{n+1} + \cdots + \phi_p y_{n+3-p} + \epsilon_{n+3} \mid y_1, \ldots, y_n, \theta) \\
&= \text{var}(\phi_1 y_{n+2} + \phi_2 y_{n+2} + \epsilon_{n+3} \mid y_1, \ldots, y_n, \theta) \\
&= \text{var}(\phi_1 y_{n+2} + \phi_2 y_{n+2} \mid y_1, \ldots, y_n, \theta) + \text{var}(\epsilon_{n+3} \mid y_1, \ldots, y_n, \theta).
\end{aligned}$$

The first term in the right hand side above cannot be written down in terms of $V_1(\theta)$ a nd $V_2(\theta)$ alone. It also involves the covariance between y_{n+1} and y_{n+2} (given $y_1, \ldots, y_n, \theta$).

Instead of working with the variances alone, we will get the recursion by working with the conditional **covariance matrices** of $y_{n+1}, \ldots, y_{n+k}$ (given θ and the data) for $k = 1, 2, \ldots$.

Below we review some basic facts about covariance matrices.

3 Covariance Matrices

A finite number of random variables can be viewed together as a random vector. More precisely, a random vector is a vector whose entries are random variables. Let $Y = (Y_1, \ldots, Y_n)^T$ be an $n \times 1$ random vector. Its Expectation $\mathbb{E}Y$ is defined as a vector whose ith entry is the expectation of Y_i i.e., $\mathbb{E}Y = (\mathbb{E}Y_1, \mathbb{E}Y_2, \ldots, \mathbb{E}Y_n)^T$. The covariance matrix of Y, denoted by $\text{Cov}(Y)$, is an $n \times n$ matrix whose (i, j)th entry is the covariance between Y_i and Y_j. Two important but easy facts about $\text{Cov}(Y)$ are:

1. The diagonal entries of $\text{Cov}(Y)$ are the variances of $Y_1, \ldots, Y_n$. More specifically the (i, i)th entry of the matrix $\text{Cov}(Y)$ equals $var(Y_i)$.

2. $\text{Cov}(Y)$ is a symmetric matrix i.e., the (i, j)th entry of $\text{Cov}(Y)$ equals the (j, i) entry. This follows because $\text{Cov}(Y_i, Y_j) = \text{Cov}(Y_j, Y_i)$.

The following formulae are very important:

1. $\mathbb{E}(AY + c) = A\mathbb{E}(Y) + c$ for every deterministic matrix A and every deterministic vector c.

2. $\text{Cov}(AY + c) = A\text{Cov}(Y)A^T$ for every deterministic matrix A and every deterministic vector c.

As a consequence of the second formula above, we get

$$\text{var}(a^T Y) = a^T \text{Cov}(Y) a = \sum_{i,j} a_i a_j \text{Cov}(Y_i, Y_j) \qquad \text{for every } p \times 1 \text{ vector } a.$$

Given two random vectors Y $(p \times 1)$ and W $(q \times 1)$, we use $\text{Cov}(Y, W)$ to denote the $p \times q$ matrix whose $(i, j)^{th}$ entry equals the covariance $\text{Cov}(Y_i, W_j)$ between Y_i and W_j. With this definition, the previous notion of $\text{Cov}(Y)$ equals simply $\text{Cov}(Y, Y)$. It can be checked that

$$\text{Cov}(AY + c, BW + d) = A\text{Cov}(Y, W)B^T.$$

4 Covariance Recursion in $AR(p)$

We shall set up a recursion for the covariance matrices:

$$\Gamma_k(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k} \end{pmatrix} \mid \theta, y_1, \ldots, y_n\right)$$

The $(i,j)^{th}$ entry of $\Gamma_k(\theta)$ is

$$\text{Cov}\left(y_{n+i}, y_{n+j} \mid y_1, \ldots, y_n, \theta\right).$$

The diagonal entries of $\Gamma_k(\theta)$ equal $V_1(\theta), \ldots, V_k(\theta)$.

To initialize the recursion for $\Gamma_k(\theta)$, note that

$$\begin{aligned}
\Gamma_1(\theta) &= \text{var}\left(y_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\
&= \text{var}\left(\phi_0 + \phi_1 y_n + \ldots \phi_p y_{n+1-p} + \epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\
&= \text{var}\left(\epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) = \sigma^2
\end{aligned}$$

We now relate $\Gamma_{k+1}(\theta)$ to $\Gamma_k(\theta)$ to establish the recursion. We can write

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix}$$

where

$$\gamma_{k1}(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n\right)$$

and, as before,

$$V_k(\theta) = \text{var}\left(y_{n+k} \mid \theta, y_1, \ldots, y_n\right).$$

We compute $\hat{\gamma}_{k1}$ as

$$\begin{aligned}
\gamma_{k1}(\theta) &:= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data}\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data}\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data}\right)
\end{aligned}$$

where, for $i = 1, \ldots, k-1$,

$$a_i = \begin{cases} \phi_{k-i} & \text{provided } k-p \le i \le k-1 \\ 0 & \text{otherwise} \end{cases}$$

Thus if a is the $(k-1) \times 1$ vector with entries $a_1, \ldots, a_{k-1}$, we have

$$\gamma_{k1}(\theta) = \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, a^T \begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix} \mid \theta, \text{data}\right) = \Gamma_{k-1}(\theta) a.$$

Further

$$\begin{aligned}
V_k(\theta) &= \text{var}\left(y_{n+k} \mid \theta, \text{data}\right) \\
&= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data}\right) \\
&= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data}\right) + \sigma^2 \\
&= \text{var}\left(\sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data}\right) + \sigma^2 = a^T \Gamma_{k-1}(\theta) a + \sigma^2.
\end{aligned}$$

The equation for obtaining $\Gamma_k(\theta)$ from $\Gamma_{k-1}(\theta)$ is therefore

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix} = \begin{pmatrix} \Gamma_{k-1}(\theta) & \Gamma_{k-1}(\theta) a \\ a^T \Gamma_{k-1}(\theta) & a^T \Gamma_{k-1}(\theta) a + \sigma^2 \end{pmatrix}$$

The algorithm for calculating the variances $V_i(\theta)$ for $i = 1, 2, \ldots, K$ is thus given by

1. Initialize with $\Gamma_1(\theta) = V_1(\theta) = \sigma^2$.

2. For $k = 2, 3, \ldots, K$, repeat the following

   a) Form the $(k-1) \times 1$ vector a whose i^{th} entry is ϕ_{k-i} if $k - p \leq i \leq k-1$ and 0 otherwise.

   b) Calculate $\Gamma_k(\theta)$ using $\Gamma_{k-1}(\theta)$ and a by the formula given above.

3. The variances $V_i(\theta), i = 1, 2, \ldots, K$ are given by the diagonal entries of the matrix $\Gamma_K(\theta)$.

Because θ is unknown, in practice, we run this recursion with θ replaced by its conditional MLE $\hat{\theta}$. The prediction standard errors are then the square roots of $V_i(\hat{\theta})$. These prediction errors coincide with those given by the `get_prediction` function in `statsmodels`.

5 AR Residuals and Autocorrelation

In the AR model, we assume $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$. One can check if a different model is apropriate for ϵ_t by examining the residuals from the fitted AR model. Specifically, let

$$\hat{\epsilon}_t := y_t - \hat{\phi}_0 - \hat{\phi}_1 y_{t-1} - \cdots - \hat{\phi}_p y_{t-p}$$

be the residuals corresponding to the fitted AR(p) model. A standard diagnostic tool in time series is to look at the AutoCorrelation Function (ACF) of the residuals. This will indicate if there are correlations between the residuals leading to violation of the i.i.d $N(0, \sigma^2)$ model (and suggesting some alternative models).

The AutoCorrelation Function is defined as follows. Consider a time series dataset $z_1, \ldots, z_n$. The AutoCorrelation function of this dataset at a lag $h \geq 0$ is denoted by r_h and it is defined as a simpler version of the correlation between (z_t, z_{t+h}) for $t = 1, \ldots, n-h$. This correlation is given by:

$$\frac{\sum_{t=1}^{n-h}(z_t - \bar{z}^{(1)})(z_{t+h} - \bar{z}^{(2)})}{\sqrt{\sum_{t=1}^{n-h}(z_t - \bar{z}^{(1)})^2}\sqrt{\sum_{t=1}^{n-h}(z_{t+h} - \bar{z}^{(2)})^2}} \quad \text{where } \bar{z}^{(1)} = \frac{1}{n-h}\sum_{t=1}^{n-h} z_t \quad \text{and} \quad \bar{z}^{(2)} = \frac{1}{n-h}\sum_{t=1}^{n-h} z_{t+h}.$$

When $n - h$ is large, we would usually have

$$\bar{z}^{(1)} \approx \bar{z}^{(2)} \approx \bar{z} = \frac{1}{n}\sum_{t=1}^{n} z_t \quad \text{and} \quad \sum_{t=1}^{n-h}(z_t - \bar{z}^{(1)})^2 \approx \sum_{t=1}^{n-h}(z_{t+h} - \bar{z}^{(2)})^2 \approx \sum_{t=1}^{n}(z_t - \bar{z})^2.$$

These approximations lead to the following simpler version of the formula for correlation between $(z_t, z_{t+h}), t = 1, \ldots, n-h$:

$$r_h := \frac{\sum_{t=1}^{n-h}(z_t - \bar{z})(z_{t+h} - \bar{z})}{\sum_{t=1}^{n}(z_t - \bar{z})^2}.$$

r_h will be called the (sample) autocorrelation at lag h. All these autocorrelations $r_1, r_2, r_3, \ldots, r_{h_{\max}}$ for some $h_{\max}$ will be collectively referred to as the AutoCorrelation Function (ACF) of the data $z_1, \ldots, z_n$. Note that $r_0 = 1$.

When $z_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$, we would expect each r_h to be small. If one or more r_h-values are large, then there is indication that $z_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$ is not a good model for the observed $z_1, \ldots, z_n$.

After fitting an AR(p) model, it is standard practice to plot the AutoCorrelation Function of the residuals to see if any of the autocorrelations take large values. If they do, we can try to replace the model $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$ (for the errors in AR(p)) by a alternative modeling assumption which allows for dependence between ϵ_t's. We shall look at such modeling assumptions next week. Moving Average (MA) are simple models which allow for dependence between 'nearby' ϵ_t's. The simplest MA model is:

$$\epsilon_t = u_t + \theta u_{t-1} \qquad \text{where } u_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2) \tag{3}$$

where θ denotes an unknown parameter. For this model clearly $\epsilon_t = u_t + \theta u_{t-1}$ and $\epsilon_{t+1} = u_{t+1} + \theta u_t$ have correlation because both involve u_t. The model (3) is called MA(1) (Moving Average Model of order 1). More generally, there are MA(q) models for $q \geq 1$ which we shall look at next week.

If we use an AR(p) model but model ϵ_t as MA(q) (instead of i.i.d $N(0, \sigma^2)$), we obtain the ARMA(p, q) model. We shall discuss these in detain next week.